Public Health Agency of Canada
Agence de la santé publique du Canada

DATA STRATEGY

This document is intended for internal use by the Public Health Agency of Canada to guide data modernization efforts and is not intended for public distribution.

Canada

We are pleased to introduce PHAC's renewed Data Strategy for 2025/26–2027/28.

Modern, secure, and timely health data are foundational to our ability to safeguard and improve the health of all people in Canada. More than just numbers, public health data tell the stories of individuals and communities. When responsibly and transparently collected, shared, and used, data provide the insights needed to guide action, ensure equitable health outcomes, and build a resilient and responsive public health system.

The COVID-19 pandemic brought these issues into sharp focus. It underscored the long-standing challenges in health data collection, sharing, and use, while also demonstrating the vital importance of cross-jurisdictional partnerships, modern tools, and coordinated action. Canadians expect us to lead with integrity and collaboration. This means strengthening governance, building trust through transparent practices, and aligning with partners across governments, Indigenous communities, and the broader health ecosystem.

At the heart of this Strategy is a strong commitment to equity. Disaggregated data, Indigenous data sovereignty, and accessible information empower us to better understand and respond to the diverse health needs of populations across Canada, ensuring no one is left behind.

This Strategy supports PHAC's role as a leader within Canada's health data ecosystem. It sets out practical actions that will strengthen data governance, improve interoperability, and foster a culture of data stewardship and literacy across the Agency. These efforts will enable more timely detection and monitoring of health issues, support innovative research, and inform public health programs and policies.

Success will depend on our collective commitment to modernizing how we manage, share, and use data as well as our ongoing collaboration and support to partners in advancing implementation. With the right tools, partnerships, and a shared vision, we can build a more equitable, effective, and trusted public health system that delivers real impact for people and communities across Canada.

Together, with data, collaboration, and decisive action, we can build a healthier future.

Nancy Hamzawi
President, Public Health Agency of Canada

Dr. Theresa Tam
Chief Public Health Officer

FOREWORD

INDIGENOUS DATA SOVEREIGNTY

Indigenous data sovereignty describes the fundamental rights of Indigenous Peoples to control, access, interpret, manage, and collectively own data about their communities, lands, and cultures. The foundations for Indigenous data sovereignty rest on Indigenous Peoples' inherent rights to autonomous governance. These are supported by global human rights agreements such as the United Nations Declaration on the Rights of Indigenous Peoples, as well as broad and regional frameworks such as the CARE Principles for Indigenous Data Governance (Collective Benefit, Authority to Control, Responsibility, Ethics) and First Nations Principles of OCAP® (Ownership, Control, Access, and Possession).

We respectfully acknowledge that the lands on which this Strategy was developed are the traditional territories of First Nations, Inuit, and Métis (FNIM) nations and communities. We recognize the rich diversity of Indigenous communities, and in our commitment to Indigenous data sovereignty, we do not seek to impose a singular vision for self-determination.

Moving forward, the Public Health Agency of Canada (PHAC, the Agency) is committed to meaningful engagement and co-development with FNIM communities. This includes creating the space and time to foster long-term relationships, ensuring that Indigenous knowledge is not siloed but rather integrated respectfully into the broader public health landscape through ongoing collaboration and partnership.

DATA AND BIAS

All data are collected and analyzed within the structures and power dynamics of the society around it. All data have bias, and, without reflection, will perpetuate existing inequities and disadvantage marginalized populations. We continue to promote health equity in data activities by promoting the application of Sex and Gender-Based Analysis Plus (SGBA+) and the advancement of anti-racism in science principles to develop, implement, and evaluate the Health Portfolio's research, surveillance, legislation, policies, regulations, programs, services, and other initiatives. The Agency's Anti-Racism in Science Action Plan and the Call to Action on Anti-Racism, Equity, and Inclusion in the Federal Public Service are also key foundations for this Strategy.

A key element of our work involves promoting the collection and use of disaggregated data, which allows for more nuanced, intersectional analysis and ensures the diverse experiences of all people in Canada are reflected. This also contributes to addressing systemic racism and other forms of discrimination to create a more equitable Canada.

We are committed to advancing equity in our data practices and addressing biases in data collection and analysis, while also amplifying the strengths, resilience, and assets of diverse communities. This includes ensuring that our data practices not only avoid perpetuating historical inequities, but also support fair, inclusive, and effective public health solutions that build on community knowledge and capacities.

INTRODUCTION

CHALLENGE

The COVID-19 pandemic highlighted persistent challenges in collecting, sharing, accessing, and using health data to benefit all people in Canada. At the same time, it served as a catalyst for significantly re-shaping how people interact with and access public health information and services.

Accurate, representative, and timely data are crucial for decision-making, providing advice during public health events (e.g., ongoing toxic drug crisis), informing routine public health practices, and taking proactive measures to prevent future events. These data guide sound public health investments, intersectoral collaboration, and equitable access to programs, contributing to improved health outcomes and health system quality, safety, and effectiveness.

The Agency's approach to collecting, protecting, sharing, and using health data is central to its mission of improving the health of all people and communities in Canada, and building public trust. Despite ongoing efforts towards data modernization, challenges such as inconsistencies with data sharing, varying data standards, and limited integration hinder the Agency's efficiency, flexibility, and responsiveness in a dynamic public health environment.

OBJECTIVE

This renewed Strategy aims to improve the timeliness, security, and efficiency of **health data*** collection, sharing, and use. With a strong focus on making the best use of limited resources to improve the efficiency of the Agency, the Strategy will also enable the Agency to strengthen horizontal collaboration and alignment with partners, promote health equity across Canada, and maximize the value of public health investments.

VISION

The Agency will be a forward-thinking partner and leader within the health data ecosystem, enabling timely, comprehensive, and interoperable data systems that can withstand evolving public health needs, inform actionable public health insights, and ultimately, improve the health for all people and communities in Canada.

* While other types of data, such as corporate reports, financial and human resources (HR) information, etc., exist within the Agency, this Strategy focuses on **health data**, which refers to observations, facts, or measurements (e.g., from surveillance and scientific research) – captured for possible further analysis, calculation or reasoning – which relate to the physical or mental health status of individuals, and socio-economic, community, and health system characteristics.

THE HEALTH DATA ECOSYSTEM

The Agency, together with other government departments, is working with provinces, territories, Indigenous leaders, pan-Canadian health organizations, and other health data partners to break down silos and modernize Canada's health system through investments for standardized health data and digital tools. This shared health priority is part of the Working Together to Improve Health Care for Canadians Plan, where federal, provincial, and territorial (FPT) governments (excluding Québec) have committed to improving how health data is collected, shared, used, and reported to manage public health events and improve health outcomes for all people in Canada. The Agency, alongside Health Canada, continues to work with partners to advance FPT efforts toward data modernization.

In line with this Plan and guided by this Strategy, the Agency must not only modernize data sharing agreements but also transform itself to be a more effective element of this ecosystem. The Agency needs to build interoperability into its core business, align health data standards and approaches, and contribute to a secure, trusted public health system of systems. This will not only support collective commitments to improve how data are collected, shared, and used, but also support actionable public health insights and inform future investments.

THE AGENCY'S DATA LANDSCAPE

Effective public health work gathers data from various sources (e.g., provincial/territorial health data systems, surveys, and public engagement) and transforms these data into evidence and information. This work then leads to evidence-informed decision-making, tailored public health messaging and reporting, and effective preparation and response to public health events.

Detecting and responding to public health threats is a shared responsibility. The Agency operates over 50 active surveillance systems, relying on over 180 sources from a variety of data providers (e.g., other federal departments, other levels of government, hospitals, etc.). Among these systems, the most frequently reported data providers were provinces/territories (64%), followed by hospitals or other healthcare settings (28%).*

To keep pace with partners and strengthen its national role and public health leadership functions, the Agency must focus on internal foundational work in data management and modernization alongside work within the broader health data ecosystem. Progress in these areas depends on close partnership with Health Canada's Digital Transformation Branch (HC-DTB), which leads information technology services under the shared corporate services model.

* Results from Data Modernization and Interoperability Initiative/Mapping questionnaires as of March 2024.

CONTEXT

GUIDING PRINCIPLES

Aligned with the 2023–2026 Data Strategy for the Federal Public Service, Roadmap 2030, and the Pan-Canadian Health Data Charter, six principles guide this Strategy:

- Purposeful
- People-focused
- Trusting
- Ethical
- Open
- Enabling

See **Appendix 2** for further details.

ENGAGEMENT AND CONSULTATION

The development of this Strategy was informed by engagement across the Agency's branches and programs, as well as previous survey and consultation findings from other initiatives. Four targeted consultations and a comprehensive survey were also conducted to validate and identify any additional key data challenges to build consensus on opportunities for improvement.

HEALTH EQUITY

The Agency strives towards health equity as both a commitment and a foundational pillar of public health. We are committed to incorporating health equity considerations within the Agency's data activities, and in turn, ensuring that the outcomes of these activities will have equitable impacts.

See **Appendix 3** for further details.

AGENCY ALIGNMENT

The Strategy is closely aligned with other existing Agency initiatives, including:

- Vision 2030: Moving Data to Public Health Action
- Strategic Surveillance Review and Plan
- Science Strategy 2024-2025 to 2029-2030
- Anti-Racism in Science Action Plan
- Open Science Action Plan
- Roadmap 2030: Excellence Through Action
- PHAC Renewal

The Strategy also supports the Agency's role in modernizing the health data ecosystem, aligning with FPT initiatives, including:

- Working Together to Improve Health Care for Canadians Plan
- Joint FPT Action Plan on Health Data and Digital Health
- FPT Data Sharing Agreement Modernization

See **Appendix 4** for further details.

PRIORITIES & KEY ACTIONS

PRIORITIES

Over the next three years, the Agency will focus its data modernization work under four key priorities. The priorities reflect the reality that the Agency, as a generator and user of health data, must collaborate with health data partners and optimize existing resources to achieve success. These cross-cutting commitments and actions will help support the vision of transforming Canada’s public health system of systems and provide timely insights for actions that improve health and reduce inequalities for all people in Canada.

1. Establish Effective Data Governance
2. Connect Data and Systems
3. Enhance Culture of Data Stewardship
4. Foster a Data-literate Workforce

PRIORITY 1.
ESTABLISH EFFECTIVE DATA GOVERNANCE

Current State. Decision making processes within the Agency are not clearly defined and data roles and responsibilities are not clearly delineated, which poses a significant risk of causing confusion, delays, and a lack of integration, oversight and accountability. This can also make it difficult to promote data stewardship and leverage data for broader organizational insights and strategic decision-making.

Goal. Improved and strengthened Agency governance structures support effective and transparent decision making on data-driven issues (e.g., data sharing, interoperability, disaggregated data, technology and artificial intelligence [AI] integration). This strengthened governance builds trust, ensures accountability, and prioritizes modernization investments, maximizing the value of the Agency's program activities. It also helps advance and enable broader FPT initiatives within the FPT health data governance structure.

1. Define clear roles and responsibilities for data management, data services and tools, and priority data sets across the Agency and with HC-DTB in alignment with data ownership and governance of health data partners. This will also help improve data accessibility and identify areas of collaboration towards shared public health priorities.
2. Strengthen an integrated data governance structure and establish a clear data governance pathway, including a new Surveillance Technical Advisory Group, to promote horizontal and technical alignment of data management and surveillance initiatives, tools, and investments.
3. Collaborate with HC-DTB to ensure a public-health focused data architecture that enables flexible data systems, scalable solutions, and timely data capability that respond to partner needs and limitations.
4. Implement the merging of the Chief Data Officer and Chief Scientific Data Officer responsibilities to ensure consistent consideration for science, research, and data.

PRIORITY 2.
CONNECT DATA AND SYSTEMS

Current State. While some examples of integrated data systems and approaches exist, there are other areas where fragmentation of data systems continue with varying data transfer methods, standards, formats, and storage solutions. This creates Agency risk, locks data value away, and prevents appropriate integration that supports the assessment of program impacts and public health outcomes.

Goal. Data systems are modernized and interoperable-by-design whenever appropriate. They incorporate pan-Canadian interoperable health data standards and new technologies to drive advanced analytics and maximize the value of data assets for public health insights that meet end-user and program needs. High-quality data, including disaggregated data when applicable, are available for analysis on demand across Agency boundaries, while ensuring security, privacy, and responsible use.

1. Align Agency data systems, agreements, and investments with evolving disaggregated data needs, standards, and guidance to support SGBA+, anti-racism, and equity-informed analysis when and where applicable.
2. Align Agency data models and systems with international and national data standards, including those developed and adopted through the shared Pan-Canadian Interoperability Roadmap with Canada Health Infoway and the Canadian Institute for Health Information (CIHI).
3. Pilot and expand a simplified data submission process for centralized, flexible intake of data from external sources (e.g., provinces/territories) to streamline data submission and management processes (e.g., harmonization of data inputs), enhance workflow efficiency, and facilitate interoperability.
4. Advance data management and interoperability capacity and utilize established platforms (e.g., CEDARS, ITAP) and proven, cost-effective tools (e.g., Django, R, Python) where applicable to support scalable and interoperable-by-design solutions.
5. Advance data linkage activities (e.g., laboratory and epidemiologic data, survey data) by leveraging existing linkages and exploring operational models for building data linkage infrastructure (e.g., models or standards from Statistics Canada and CIHI).
6. Collaborate with HC-DTB to ensure data management practices are embedded in solutions to enable interoperability and reuse of data for Agency program and partner needs.

PRIORITY 3.
ENHANCE CULTURE OF DATA STEWARDSHIP

Current State. Siloed data-custodian model practices, inconsistent standards, and limited resources hinder effective data management and integration. Most data are only accessed and used to achieve program specific outcomes and limit data discoverability and accessibility, leading to a higher risk of inefficiencies and duplication of work.

Goal. Enhanced Agency culture of shared, responsible data exchange builds trust and supports equitable public health outcomes in alignment with internal and pan-Canadian data stewardship models (while respecting FNIM models), policies, and approaches. Data stewardship principles are promoted, and stewards ensure that valuable data resources are findable, accessible, interoperable, and reusable (FAIR) across the Agency and beyond. Disaggregated data are created and used where appropriate, and data management capacity across the lifecycle (e.g., collection, use, and disposal) is prioritized.

1. Promote open science and open data practices that enhance the use of FAIR and CARE principles, and embed clauses for the secondary use of data and disaggregated data into our data sharing agreements, as appropriate, while continuing to safeguard privacy.
2. Establish a reporting and review process and dashboard for Minimum Performance Standards, including SGBA+, equity, and co-developed Indigenous data elements as available.
3. Develop Agency-wide approaches to improve awareness and incentivize open science solutions and open data publishing to enable the sharing of knowledge and timely insights.
4. Develop and/or update policies, guidelines, and frameworks and build mechanisms that enable shared and responsible data management and exchange for data stewards and data users (e.g., AI implementation framework, data management directive, guidance on privacy protection and cybersecurity in health data sharing).
5. Ensure data standards and approaches are effectively communicated and engage in open discussions within the Agency and with national-level data custodians and PT partners. This will be done through appropriate tables and channels to promote the adoption of international and national data standards.

PRIORITY 4.
FOSTER A DATA-LITERATE WORKFORCE

Current State. The Agency's workforce is technically skilled and diverse; however, it has limited availability and access to modern tools, resources, and training. This hinders workforce capacity building in evolving areas such as data literacy and analytics. The Agency must ensure our people have the modern skills and tools, so our work is as cost-efficient and effective as possible.

Goal. The Agency's workforce, across disciplines and levels, is equipped with the skills and tools to adapt to technological advancements that drive appropriate health data systems integration, informed decision-making, and innovation within the public health context. The workforce is familiar with principles of health equity, SGBA+, and data disaggregation and interpretation. The workforce also has appropriate competencies and training in alignment with the Government of Canada (GC) Data Competency Framework. Cost-effective and open data tools are used broadly, and open and common GC training is accessible.

1. Integrate data literacy training and career development in data science, epidemiology, etc. into HR planning, succession planning, and performance/talent management agreements to align and formalize training and maintain organizational stability.
2. Identify and promote the use of common data tools across the Agency to enhance collaboration, enable consistency, and help prioritize training efforts.
3. Promote and build on existing formal and informal data learning opportunities and resources (e.g., Chief Data Officer portals in the Agency and Health Canada, Communities of Practice, Synergy Series, repository for best practices, Canada School of Public Service training).
4. Strengthen talent mobility practices through micro-missions, rotational assignments, etc. to encourage inter-program learning/cross-training within the Agency.
5. Build talent and expand capacity by recruiting students with cutting-edge skills in data, AI, and other specialty areas.
6. Create a change management strategy, including a change readiness assessment and engagement and communication plan, to foster data literacy and a culture of collaboration and innovation.
7. Advance anti-racism in science and incorporate principles and approaches in workforce activities to support equity, diversity, and inclusion and promote reporting of equity employment data.

AFTERWORD

People in Canada relied on the Public Health Agency of Canada during the unprecedented emergency of the COVID-19 pandemic. The Agency built new capabilities to protect the public from health threats by gathering better surveillance data, integrating our systems, and more clearly articulating priorities, roles and responsibilities. Our work is far from done. We have a responsibility to learn and integrate these lessons so that we can appropriately respond to ongoing and future threats to population health.

The Agency is a science-based organization that relies on data and the experts that use it. Our mission drives us to collect information and detect signals about what is happening in the world around us. Our experts analyze the information and work with data ecosystem partners to take the most effective actions to protect and promote the health of all people in Canada.

The priorities and actions articulated in this Strategy are important, achievable and cost-effective ways to help us achieve these objectives in this rapidly evolving environment. Success will depend on strong partnerships between scientists, experts, and technology enablers across the Agency and beyond.

In a post-COVID world, data stewardship must continue to evolve beyond just safeguarding data.

NEW & FUTURE TECHNOLOGIES: **ARTIFICIAL INTELLIGENCE**

The Agency must strengthen its core data practices and systems and formalize AI best practices in our data governance structures to efficiently and responsibly leverage these technologies for public health action. Focusing on data quality and interoperability ensures that the potentially transformative use of AI can be scaled, and this sets the stage for the Agency to use emerging technologies to empower public health professionals and further protect the health of people in Canada.

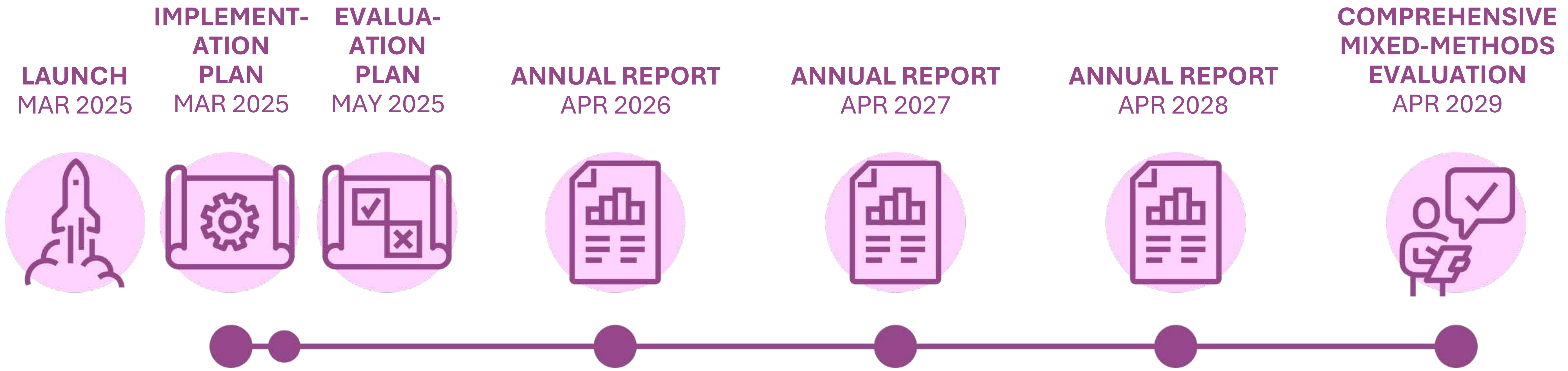
We must take meaningful action to advance our efforts towards health equity, addressing biases in data and promoting the use of disaggregated data. By working as part of an integrated ecosystem, we can ensure that data are accessible by partners and create the most value when it is most needed.

It is my honour to work alongside so many talented and dedicated public health professionals. I am confident the Agency is ready to play its crucial role in working with colleagues and partners to realize a world-class health data system for Canada.

Alyssa Daku
Chief Data Officer

APPENDIX 1.
EVALUATION

A multi-pronged approach will be used to monitor and evaluate the implementation of the Data Strategy. Within a year of publishing PHAC's Data Strategy, a process evaluation will be conducted to assess progress. The Office of Audit and Evaluation will be consulted to establish key metrics for each public health mission area and develop a measurement plan. These metrics will be monitored and reported annually. A mixed methods evaluation of effectiveness will be conducted after four years to assess the effectiveness of the four public-health mission areas.



APPENDIX 2.
GUIDING PRINCIPLES

Aligned with the 2023–2026 Data Strategy for the Federal Public Service, Roadmap 2030, and the Pan-Canadian Health Data Charter, six principles guide this strategy.

Purposeful. Data collection, sharing and use should have clear, defined goals ultimately aimed at improving public health outcomes. Data should guide evidence-based decisions, support health equity, and strengthen public health systems, always aligning with the goal of improving the health of people and communities in Canada.

People-focused. Data should be focused on improving public health outcomes for individuals and communities, including those who are equity-deserving or who are in greatest need.

Trusted. Data should be reliable, secure, and protected, to build and maintain public trust in how health data are used for public health purposes. They should be protected from misuse, grounded in a strong data governance foundation.

Ethical. Data should be collected, shared, and used in a manner that respects persons and communities and promotes the public good. Data sharing, collection and use should be transparent, consider potential benefits and harms, and promote fairness, especially among groups who are marginalized, disadvantaged or vulnerable to injustice.

Open. Data should be findable and accessible, encouraging collaboration and innovation. They should foster an open environment that supports research, improved public health outcomes, and timely response, including in emergencies.

Enabling. Data should enable better decision-making, evidence-based policies, programs, services, and research. They should be supported by the provision of tools, skills, and resources needed to use data effectively including responsible data management processes.

APPENDIX 3.
ENGAGEMENT SUMMARY

COVID-19 DATA CHALLENGES

The pandemic exposed longstanding challenges in data collection, sharing, access, and use across the public health ecosystem, in Canada. These challenges highlighted the need for a more cohesive public health framework to support timely and effective responses to public health crisis.

This pandemic was a catalyst for change, emphasizing the importance of building a strong, modernized, and equity-informed data environment.

Insights gathered through the Agency-wide consultations shape the renewed strategy, ensuring it reflects the needs and experiences of the Agency's workforce while fostering collaboration and innovation in public health data management.

KEY CHALLENGES

Data Governance. Decentralized data governance and stewardship efforts result in inconsistent practices, duplication and a lack of collaboration across the Agency.

Data Access and Sharing. Complex data-sharing agreements and approval processes affect timely access to and sharing of data, as well as data disaggregation.

Interoperability. Data-sharing infrastructure lacks flexibility and increases risk of redundancies.

Data Privacy and Trust. The Agency is facing challenges in maintaining public trust. There is a need for transparent communication to bolster public confidence and address privacy concerns.

Workforce Capacity. The Agency's workforce needs structured learning paths, improved access to data tools, and professional development to enhance data management skills.

OPPORTUNITIES

Strengthening Data Governance. Establishing a Tier III Data Governance Committee at senior management level ensures alignment, reduces redundancy, and unifies surveillance efforts. Creating a central technical table with provinces and territories enhances data sharing, while creating a systems map and resource portal could enhance data stewardship and the disaggregation of data.

Advancing Data Tools & Integration. Existing tools (ITAP and CEDARS) can be expanded. Artificial intelligence can be explored for data analysis. A centralized data inventory and dashboard can advance the Agency's capabilities.

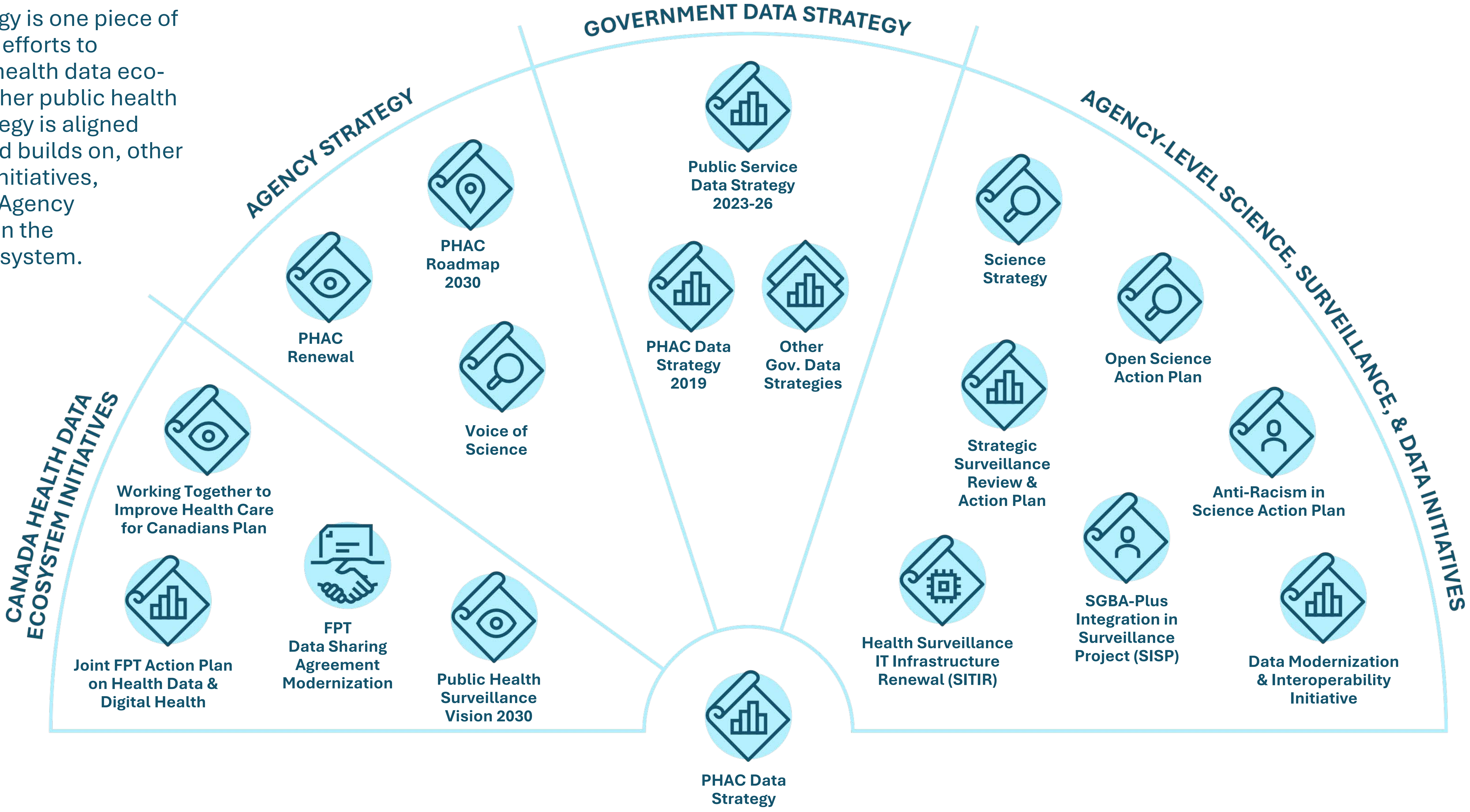
Building Interoperability. Adopt a "Single Front Door" with flexibility, prioritize modernizing legacy systems, develop a cohesive interoperability framework, prioritize inclusivity in data partnerships to promote health equity.

Data Privacy and Trust. Promote data literacy to reduce misinformation. Implement feedback channels, strengthen privacy protection, and shift to proactive transparency.

Building Workforce Capacity. Implement data literacy programs for leadership and tailored training paths for staff. Enhance access to modern data tools, and leverage external partnerships to strengthen internal data skills.

APPENDIX 4.
LINKAGES

The Data Strategy is one piece of a larger suite of efforts to modernize the health data eco-system and further public health goals. The Strategy is aligned closely with, and builds on, other strategies and initiatives, both within the Agency and elsewhere in the health data ecosystem.



APPENDIX 5.
GLOSSARY

Data Architecture. The structure of an organization's data assets that translates business needs into data and system requirements and seeks to manage data and its flow through the enterprise.

Data Custodian. An individual or organization responsible for the secure collection and/or storage of health data and the curation of health data use, disclosure, retention, and disposal; primarily concerned with security and privacy of data.

Data Governance. The exercise of authorities, control, and shared decision-making (planning, monitoring and enforcement) over the management of data assets. It guides behaviour to ensure that data is managed as an organizational asset with alignment of foundational data initiatives and the organization's mandate and objectives.

Data Literacy. The ability to derive meaningful information from data, focusing on competencies in reading, analyzing, interpreting, visualizing, and communicating data, while understanding its role in decision-making. It also includes skills in data stewardship, such as assessing data quality, securing data, and ensuring its responsible and ethical use.

Data Management. Activities related to the development, execution, and supervision of plans, policies, programs, and practices that deliver, control, protect, and enhance the value of data assets throughout its lifecycle.

Data Modernization. The continuous efforts to improve and transform the collection, sharing, analysis, and use of data, with the goal of creating modern and responsive data systems.

Data Standards. The rules and specifications by which data are described, defined, and recorded. In order to share, exchange, and understand data, standardized formats and meanings are needed.

Data Stewardship. The responsible governance and management of data to ensure its accuracy, security, and ethical use, while maximizing its value for the public good.

Disaggregated Data. Compiled or aggregate data that has been separated or broken down into smaller information units for the purposes of analysis. Disaggregated data allows for detailed analysis and insight about various subsets or outcomes within a larger data set. Data can be disaggregated by variables such as income or socio-cultural background.

Health Data. Observations, facts, or measurements – captured for possible further analysis, calculation or reasoning – which relate to the physical or mental health status of individuals, health system performance and socio-economic, community and health system characteristics.

Health Data System. The infrastructure that enables collection, access, sharing, use, and protection of health data, supported by networks of people, organizations, policies, and technologies.

Health Equity. The absence of unfair systems and policies that cause health inequalities. It seeks to reduce inequalities and increase access to opportunities and conditions conducive to health for all.

Indigenous Data Sovereignty. The concept of Indigenous authority, right, power to govern as sovereign Nations and make decisions or laws on the ownership, control, collection, access, analysis, application, possession, and use of their own data.

Interoperability. The ability to exchange, use, and interpret information across diverse systems in a seamless, secure, and timely way.

Intersectionality. A framework for understanding how overlapping systems of power and social identity factors (such as race, gender, and class) intersect and contribute to social inequalities and disparities in health outcomes.

Open Data. Structured data that is machine-readable, freely shared, used, and built on without restrictions.

Public Health Surveillance. The ongoing collection, analysis, interpretation, and dissemination of health data for the purpose of planning, implementing, and evaluating interventions to protect and improve the health of populations.

Surveillance System. A set of processes for collecting, managing, analyzing, interpreting, and disseminating data to support public health decision-making and action, supported by enabling components such as legislation, partnerships, technology, trust, and skilled workers.

System of Systems. A network of interconnected sub-systems working together toward a common goal.

Public Health Agency of Canada, 2025